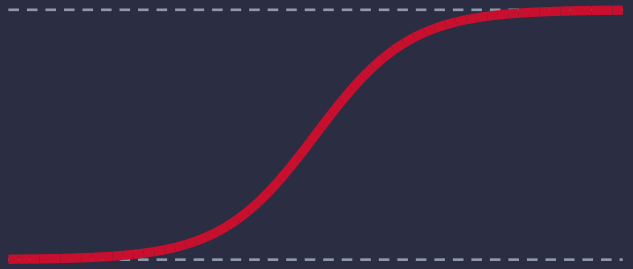
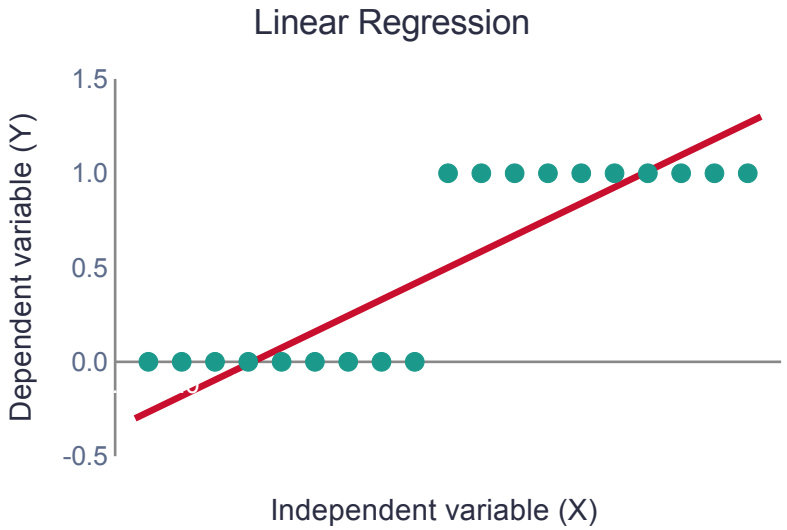
Logistic Regression

Algorithm & Solved Example



Linear vs Logistic Regression





Straight line: predicted Y can exceed the 0 and 1 range

S-curve: predicted Y lies within the 0 and 1 range

Why Logistic Regression?



Linear regression

Predicts the numerical response but is not suitable for predicting the categorical variables.



Classification problem

When categorical variables are involved, it is called a classification problem.



Binary classification

Logistic regression is suitable for binary classification problems.

Predicting Categorical Variables

For example, the following scenarios are instances of predicting categorical variables:



1

Spam or not spam?

Is the mail spam or not spam? The answer is yes or no. Thus, the categorical dependent variable is a binary response of yes or no.



2

Admitted or not?

Whether the student should be admitted or not is based on entrance examination marks. Here, the categorical variable response is admitted or not.



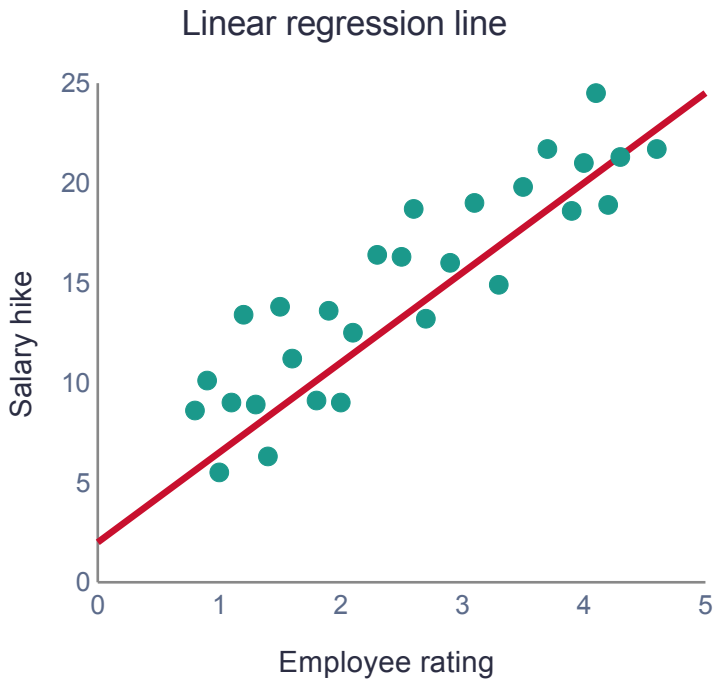
3

Pass or fail?

The student being pass or fail is based on marks secured.

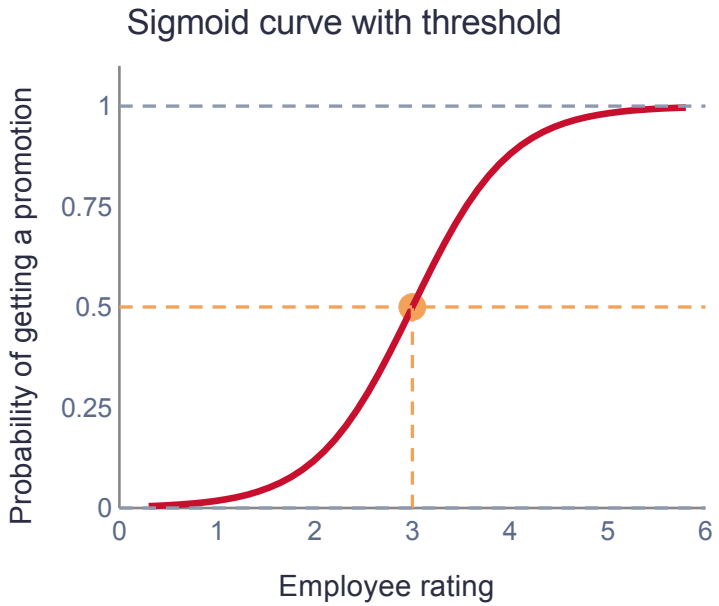
How Does the Algorithm Work?

- Consider the following example:
- An organization wants to determine an employee's salary increase based on their performance.
- For this purpose, a linear regression algorithm will help them decide.
- Plotting a regression line by considering the employee's performance as the independent variable, and the salary increase as the dependent variable will make their task easier.



From a Straight Line to an S-Curve

- Now, what if the organization wants to know whether an employee would get a promotion or not based on their performance?
- The above linear graph won't be suitable in this case.
- As such, we clip the line at zero and one, and convert it into a sigmoid curve (S curve).
- Based on the threshold values, the organization can decide whether an employee will get a salary increase or not.

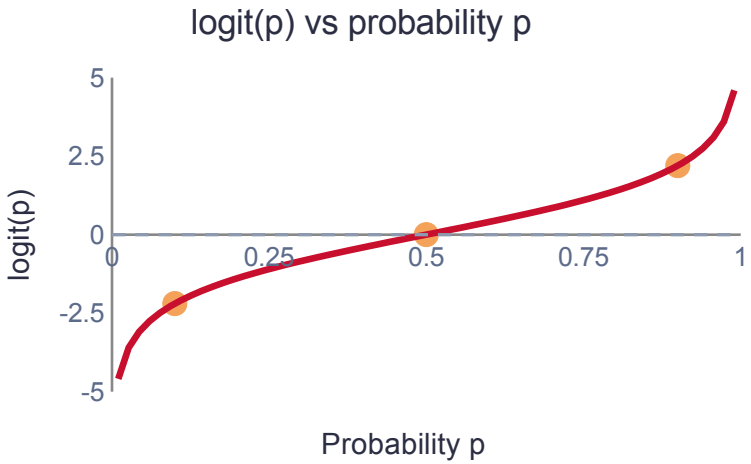


Threshold (th) = 0.5 → cut-off rating = 3

The Logit Function

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x$$

- The log of the odds is called the logit (log-odds) of p.
- Logit maps a probability between 0 and 1 to any real number from −∞ to +∞.
- The logit is linear in x, which is why logistic regression is a “regression”: we fit a straight line to the log-odds.
- The sigmoid is the inverse of the logit: it maps $\beta_0 + \beta_1 x$ back to a probability.
- β_1 is the change in log-odds for a one-unit increase in x; e^β_1 is the odds ratio.



p	Odds p/(1−p)	logit(p)
0.1	0.11	−2.20
0.5	1	0
0.9	9	+2.20

The Odds of Success

To understand logistic regression, let's go over the odds of success.

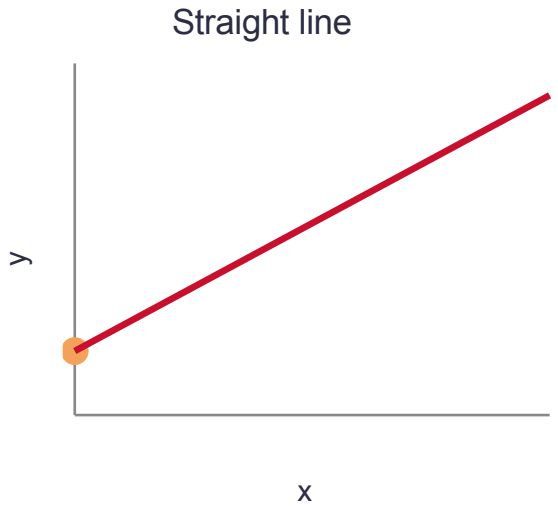
$$\text{Odds } (\theta) = \frac{\text{Probability of an event happening}}{\text{Probability of an event not happening}}$$

$$\text{Odds } (\theta) = \frac{p}{1-p}$$

The values of odds range from **zero to ∞** and the values of probability lie between **zero and one.**

Consider the equation of a straight line:

$$y = \beta_0 + \beta_1 * x$$



β_0 = intercept β_1 = slope

Taking the Log of the Odds

1 Now to predict the odds of success, we take log on the odds formula:

$$\log\left(\frac{p(x)}{1-p(x)}\right)=\beta_0+\beta_1 x$$

2 Exponentiating both the sides, we have:

$$e^{\ln\left(\frac{p(x)}{1-p(x)}\right)}=e^{\beta_0+\beta_1 x}$$

$$\downarrow$$

$$\left(\frac{p(x)}{1-p(x)}\right)=e^{\beta_0+\beta_1 x}$$

Using the identity

$$e^{\ln(x)}=x$$

Deriving the Sigmoid Function

Let $Y = e^{\beta_0 + \beta_1 * x}$

Then $\frac{p(x)}{1 - p(x)} = Y$

$p(x) = Y(1 - p(x))$

$p(x) = Y - Y(p(x))$

$p(x) + Y(p(x)) = Y$

$p(x)(1 + Y) = Y$

$p(x) = \frac{Y}{1 + Y}$

Substitute **Y = e^(β₀ + β₁x)** in p(x) = Y / (1 + Y):

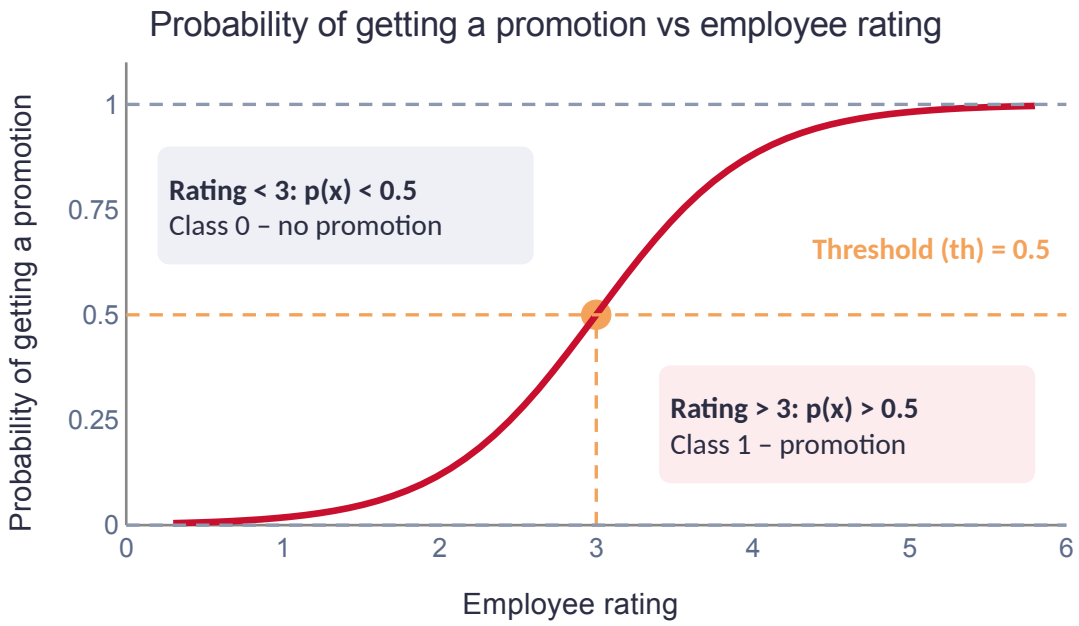
$$p(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

Dividing numerator and denominator by e^(β₀ + β₁x):

$$p(x) = \frac{1}{e^{-(\beta_0 + \beta_1 x)} + 1}$$

The Sigmoid Curve

The sigmoid curve obtained from the above equation is as follows:



$$p(x) = \frac{1}{1 + e^{-y}}$$

where $y = \beta_0 + \beta_1 x$

Output always between 0 and 1 so it can be read as a probability. The orange dashed line marks the threshold 0.5; it crosses the curve at rating 3, the decision boundary.

Understanding the Threshold

What it is

The threshold (th) is the cut-off probability that turns the sigmoid output p(x) into a class label. The usual default is 0.5.

Decision rule

If $p(x) \geq th \rightarrow$ Class 1 (promotion / selected / spam).
If $p(x) < th \rightarrow$ Class 0 (no promotion / not selected / not spam).

Decision boundary

With th = 0.5, p(x) = 0.5 exactly when $\beta_0 + \beta_1 x = 0$, i.e. $x = -\beta_0 / \beta_1$. In the promotion graph this is an employee rating of 3.

Choosing the value

A higher th (e.g. 0.7) is stricter: fewer positives and fewer false alarms. A lower th (e.g. 0.3) catches more positives but raises false alarms. Pick it based on the cost of each error.

Solved Example: Student Admission

- The student dataset has entrance marks based on the historic data of those who are selected or not selected.
- Based on the logistic regression, the values of the learnt parameters are $\beta_0 = 1$ and $\beta_1 = 8$.
- Assuming marks of x = 60, compute the resultant class.

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

$$\beta_0 + \beta_1 x = 1 + 8 \times 60 = 481$$

$$p(x) = \frac{1}{1 + e^{-(1 + 8 \times 60)}} = \frac{1}{1 + e^{-481}}$$

$$p(x) = \frac{1}{1 + e^{-481}} = 0.44$$

Substituting x = 60 into the sigmoid

Check the arithmetic: e^(−481) ≈ 0, so 1 / (1 + e^(−481)) ≈ 1.0, not 0.44. With the parameters as given, p ≈ 1 > 0.5 would mean "selected".

If we assume the threshold value as 0.5, then it is observed that 0.44 < 0.5; therefore, the candidate with marks 60 is not selected.

Linear vs Logistic Regression

Aspect	Linear Regression	Logistic Regression
Problem type	Regression – predicts a numerical response	Classification – predicts a categorical (binary) response
Output	Continuous value, e.g. salary hike	Probability p(x), e.g. chance of promotion
Output range	$-\infty$ to $+\infty$ (can exceed 0 and 1)	Always between 0 and 1
Curve	Straight line	S-shaped sigmoid curve
Equation	$y = \beta_0 + \beta_1 x$	$p(x) = 1 / (1 + e^{-(\beta_0 + \beta_1 x)})$; $\text{logit}(p) = \beta_0 + \beta_1 x$
Fitting method	Least squares (minimise squared error)	Maximum likelihood (minimise log loss)
Threshold	Not needed	Needed to assign the class (default 0.5)
Example	Salary hike from employee rating	Promotion yes/no; admitted or not